

DECODELABS Internship 2026

Project 1: Data Integrity & Change Log

- **Intern Name:** Martina Marco Dawoud
- **Track:** Data Analytics
- **Project Title:** Data Cleaning & Preparation

Changes	Column Name	Description of Issue	Action Taken (Data Cleaning)	Impact & Business Logic	Status
1	OrderID	Potential duplicate records or spaces in IDs.	Verified that all transaction IDs are unique (\$0\%\$ error rate) and applied TRIM to remove whitespace.	Ensured transaction counts are accurate and established a single source of truth.	Resolved
2	Date	Inconsistent or unstandardized date values.	Standardized all formats to the ISO standard (YYYY-MM-DD).	Guaranteed historical timeline consistency for future time-series analysis.	Resolved
3	CouponCode	Missing or blank cells (\$Null\$ values).	Imputed missing text fields by replacing blanks with 'No Coupon'.	Prevented arbitrary row deletion while preserving the full statistical dataset.	Resolved

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4	UnitPrice / TotalPrice	Text spaces or decimal format discrepancies.	Fixed formatting to display numbers cleanly with \$2\$ decimals.	Ensured mathematical correctness across currency rows.	Resolved
5	<i>Business Logic</i>	Gross vs Net revenue conflict.	Observed that TotalPrice equals (\$Quantity \times UnitPrice\$) without deducting coupon codes (SAVE10, etc.).	Documented that current revenue figures represent gross revenue before discounts.	Audited
6	<i>Net_TotalPrice (New)</i>	Need to differentiate between customers who paid full price and those who used discount coupons.	Created a new calculated column (Net_TotalPrice) applying conditional math based on CouponCode (e.g., deducting 10% for SAVE10, 15% for WINTER15, and keeping 100% for FREESHIP and No Coupon).	Allowed clear differentiation between full-price sales and discounted orders, providing an accurate metric for actual net revenue.	Resolved